

MVP Roadmap

CAKE

project based on ESP12F and ESP32 with an MQTT broker (EMQX)





MVP Development Milestones (3 Months)

1

Month 1: Orientation, Hardware, and Initial Software Development

2

Month 2: Hardware and Software Development (Continued)

3

Month 3: Software Development, Integration, and Testing

4

Post 3 Months (Beyond MVP)



Month 1: Orientation, Hardware, and Initial Software Development

Week 1–2: Orientation and Setup

- Set up the development environment.
- Familiarize yourself with KIT's IRF facilities and resources.
- Meet with assigned mentors for initial discussions.

Week 3–4: Component Planning and Procurement

- Identify and plan the required components for the smart switch.
- Create a detailed component procurement plan.

Week 4: Initial PCB Design and Programming (ESP12F and ESP32)

- Develop and program the microcontroller unit (MCU) for the smart switch.
- Ensure basic functionality and communication capabilities.
- Initiate the development of server-side components using Node.js.
- Start the initial design phase of the PCB for the smart switch.



Month 2: Hardware and Software Development (Continued)

Week 1–2: PCB Prototyping and Testing

- Prototype the initial PCB design for testing purposes.
- Verify hardware functionality and identify potential improvements.

Week 3–4: PCB Refinement and Finalization

- Refine the PCB design based on testing results.
- Finalize the hardware components and layout.

Week 4: Dashboard and Mobile App Development Kickoff

- Begin the development of a React-based web dashboard for controlling the smart switch.
- Initiate the development of a React Native mobile app for remote control.

Month 3: Software Development, Integration, and Testing

Week 1-2: Comprehensive Server and Client Development

- Continue the server-side development, adding features and ensuring stability.
- Advance the dashboard and mobile app development for enhanced functionality.

100%

Hardware Ready

Week 3-4: MQTT Protocol Integration and EMQX Broker Setup

- Implement MQTT communication protocol for seamless device communication.
- Set up and configure EMQX broker for reliable messaging.

100%

Client Portal

Week 4: Comprehensive Testing and Submission

- Integrate hardware and software components for end-to-end testing.
- Identify and address any issues through iterative testing.
- Submit the MVP for mentor review and feedback.

100%

Server architecture

MVP's success

Post 3 Months (Beyond MVP):

Collect user feedback on the MVP.



Iterate on the design and functionality based on user input.



Plan for scalability and additional features based on the MVP's success.



This adjusted plan includes the Week 1-2 orientation and setup phase, ensuring a smooth transition into the development process with the integration of both hardware and initial software development activities. Regular communication with mentors is essential for maintaining alignment with project goals and expectations. Adjustments can be made based on the progress and feedback received during the pre-incubation period.

Thank You



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